Dr. Taeyeong Choi *Curriculum vitae*

CONTACT INFORMATION

Kennesaw State University
Department of Information Technology
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INTERESTS

Real-world AI Applications; Robot/Computer Vision; Agricultural Technologies; Food Safety/Security; Anomaly Detection; Representation Learning

CURRENT ACADEMIC APPOINTMENTS

Kennesaw State University (KSU), Marietta, GA, USA
Assistant Professor **Aug 2023 – Present**

- Department of Information Technology

PREVIOUS ACADEMIC APPOINTMENTS

University of California, Davis (UCD), Davis, CA, USA
Postdoctoral Scholar **Aug 2022 – Jul 2023**

- Supervisor: Dr. Xin Liu
- Affiliations:
  - Department of Computer Science
  - AI Institute for Food Systems (AIFS)

University of Lincoln (UoL), Lincoln, UK
Postdoctoral Research Associate **Oct 2020 – Jul 2022**

- Supervisor: Dr. Grzegorz Cielniak
- Affiliations:
  - Lincoln Agri-Robotics (LAR)
  - Lincoln Institute for Agri-food Technology (LIAT)
  - Lincoln Centre for Autonomous Systems (L-CAS)

EDUCATION

Arizona State University (ASU), Tempe, AZ, USA
M.S. & Ph.D., Computer Science **Dec 2020**

- Advisor: Dr. Theodore (Ted) P. Pavlic
- Ph.D. Dissertation: "Deep Learning Approaches for Inferring Collective Macrostates from Individual Observations in Natural and Artificial Multi-Agent Systems Under Realistic Constraints". ISBN: 9798557031004

Soongsil University (SSU), Seoul, South Korea
B.S.E., Computer Science and Engineering **Aug 2015**

PUBLICATIONS

[1] (To appear) **Choi T.** and M. Kamruzzaman. Are Time-Series Foundation Models Ready for E-Nose Data? An Empirical Assessment of Their Embeddings. In: *Proceedings of the 2026 IEEE SENSORS (SENSORS 2026)*, Oct 25–Oct 28, 2026. Rotterdam, The Netherlands. Pre-print: https://arxiv.org/abs/2606.27672.

[2] (To appear) Mamo, R., R. D. Silva, G. Cielniak, and **T. Choi**. Enabling 24-hour Agricultural Robotics: Unsupervised Day-to-Night Cross-Modal Image Translation for Nighttime Visual Navigation. In: *Proceedings of the 2026 IEEE/RSJ International Conference on Intelligent Robots & Systems (IROS 2026)*, Sep 27–Oct 1, 2026. Pittsburgh, PA, USA. Pre-print: https://arxiv.org/abs/2607.12065—Finalist for the Best Paper Award on Agri-Robotics.

[3] (To appear) Supannopaj, T., N. Voore, A. Taggart, **T. Choi**, and C. Huang (2026): Tackling Catastrophic Forgetting in Split Learning via Orthogonal Projection. In: Proceedings of the 35th International Conference on Artificial Neural Networks (ICANN 2026). Sep 14–Sep 17, 2026. Padua, Italy.

[4] Mamo, R. and **T. Choi**. CA-Cut: Crop-Aligned Cutout for Data Augmentation to Learn More Robust Under-Canopy Navigation. In: *Proceedings of the 14th European Conference on Mobile Robots (ECMR 2025)*, Sep 2–Sep 5, 2025. Padua, Italy. doi:10.1109/ECMR65884.2025.11

[5] Ahmed, N., **T. Choi**, X. Zhang, S. Islam, and R. Islam. Towards Safer Mixed Reality with Auditory Warnings for Uncontrolled Indoor Settings. In: *Proceedings of 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW 2025)*, Mar 3–Mar 8, 2025. Saint Malo, France. doi:10.1109/VRW66409.2025.00071.

[6] **Choi, T.**, O. Would, A. Salazar-Gomez, X. Liu, and G. Cielniak. Channel Randomisation: Self-Supervised Representation Learning for Reliable Visual Anomaly Detection in Specialty Crops. In: *Journal of Computers and Electronics in Agriculture*, Nov 2024. doi:10.1016/j.compag.2024.109416.

[7] **Choi T.**, D. Guevara, Z. Cheng, G. Bandodkar, C. Wang, B. N. Bailey, M. Earles, and X. Liu. DAVIS-Ag: A Synthetic Plant Dataset for Prototyping Domain-Inspired Active Vision in Agricultural Robots. In: *Proceedings of the 2024 IEEE International Conference on Automation Science and Engineering (CASE 2024)*, Aug 28–Sep 1, 2024. Bari, Italy. doi:10.1109/CASE59546.2024.10711368

[8] Goyal, S., K. Sasikumar, R. Sheth, A. Seelam, **T. Choi**, and X. Liu. EnColor: Improving Visual Accessibility with a Deep Encoder-Decoder Image Corrector for Color Vision Deficient Individuals. In: *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI-24)*, Feb 20–27, 2024. Vancouver, Canada. doi:10.1609/aaai.v38i21.30382

[9] Bandodkar, G., S. Agarwal, A. K. Sughosh, S. Singh, and **T. Choi**. "Allot?" Is "A Lot!" Towards Developing More Generalized Speech Recognition System for Accessible Communication. In: *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI-24)*, Feb 20–27, 2024. Vancouver, Canada. doi:10.1609/aaai.v38i21.30381

[10] **Choi, T.**, O. Would, A. Salazar-Gomez, and G. Cielniak. Self-supervised Representation Learning for Reliable Robotic Monitoring of Fruit Anomalies. In: *Proceedings of the 2022 IEEE International Conference on Robotics and Automation (ICRA 2022)*. May 23–27, 2022. Philadelphia, USA. doi:10.1109/ICRA46639.2022.9811954.

[11] **Choi, T.**, B. Pyenson, J. Liebig, and T. P. Pavlic. Beyond Tracking: Using Deep Learning to Discover Novel Interactions in Biological Swarms. *Journal of Artificial Life and Robotics (AROB)*, Mar 2022. doi:10.1007/s10015-022-00753-y
— Extension of the *Best Paper Award* winner at *the 4th International Symposium on Swarm Behavior and Bio-Inspired Robotics 2021 (SWARM 2021)*, Jun 1–4, 2021. Kyoto, Japan. Virtual event.

[12] **Choi, T.** and G. Cielniak. Adaptive Selection of Informative Path Planning Strategies via Reinforcement Learning. In: *Proceedings of the 10th European Conference on Mobile Robots (ECMR 2021)*, Aug 31–Sep 3, 2021. Bonn, Germany. Virtual event. doi:10.1109/ECMR50962.2021.9568796

[13] **Choi, T.**, B. Pyenson, J. Liebig, and T. P. Pavlic. Identification of Abnormal States in Videos of Ants Undergoing Social Phase Change. In: *Proceedings of the 35th AAAI Conference on Artificial Intelligence (AAAI-21)*, Feb 2–9, 2021. Virtual event. doi:10.1609/aaai.v35i17.17794

[14] **Choi, T.** and T. P. Pavlic. Automatic Discovery of Motion Patterns that Improve Learning Rate in Communication-Limited Multi-Robot Systems. In: *Proceedings of the 2020 IEEE International Conference on Multisensor Fusion and Integration (MFI 2020)*, Sep 14–16, 2020. Karlsruhe, Germany. Virtual event. doi:10.1109/MFI49285.2020.9235218

[15] Kang, S., **T. Choi** and T. P. Pavlic. How Far Should I Watch? Quantifying the Effect of Various Observational Capabilities on Long-range Situational Awareness in Multi-robot Teams. In: *Proceedings of the 1st IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2020)*, Aug 17–21, 2020. Washington, DC, USA. Virtual event. doi:10.1109/ACSOS49614.2020.00036

[16] **Choi, T.**, S. Kang, and T. P. Pavlic. Learning Local Behavioral Sequences to Better Infer Non-local Properties in Real Multi-robot Systems. In: *Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA 2020)*, May 31–June 4, 2020. Paris, France. Virtual event. doi:10.1109/ICRA40945.2020.9196728

[17] **Choi, T.**, T. P. Pavlic, and A. W. Richa. Automated Synthesis of Scalable Algorithms for Inferring Non-Local Properties to Assist in Multi-Robot Teaming. In: *Proceedings of the 2017 IEEE International Conference on Automation Science and Engineering (CASE 2017)*, Aug 20–23, 2017. Xi'an, China. doi:10.1109/COASE.2017.8256320

[18] **Choi, T.** and H. Na. Stealthy Behavior Simulations based on Cognitive Data. *Journal of Korea Game Society (JKGS)*, 16(2):27–40, Apr 2016. doi:10.7583/JKGS.2016.16.2.27

[19] **Choi, T.** and H. Na. Making Levels More Challenging with a Cooperative Strategy of Ghosts in Pac-Man. *Journal of Korea Game Society (JKGS)*, 15(5):89–98, Oct 2015. doi:10.7583/JKGS.2015.15.5.89

[20] **Choi, T.** and H. Na. Stealthy Behavior Simulations based on Cognitive Data. In: *Proceedings of the 2015 IEEE International Conference on Machine Learning and Cybernetics (ICMLC 2015)*, 16(2):27–40, Jul 12–15 2015. Guangzhou, China. doi:10.1109/ICMLC.2015.7340900

Invited Talks

[21] Self-supervised Learning of Visual Anomalies in Strawberries. In: *International Conference on Digital Technologies for Sustainable Crop Production (DIGICROP)*, Mar 2022. Virtual event.

[22] AI Research in Agriculture and Beyond – Successful Machine Learning under Limited Resources. In: *Hankyong National University, School of Computer Engineering & Applied Mathematics Seminar*, Dec 2021. Anseong, South Korea.

[23] Local Behavior Learning for Social Temperature Prediction without Individual Ant Tracking. In: Oral Session at *Collective Information Processing*, Mar 2020. Berlin, Germany.

[24] Machine Learning Applications for Video Game Development. In: *Hankyong National University, School of Computer Engineering & Applied Mathematics Seminar*, Jul 2015. Anseong, South Korea.

External Grants

NSF

- PI, "ERI: Odor Representation Learning for Robust Detection of Co-present Foods and Pathogen Contamination Using an Electronic Nose", $199,995, Jul 2025–Jun 2027.

USDA

- Co-PI, "Dynamic Time-Series and AI-Powered Spectral-Robotic Intelligence System for High-Throughput In-Ovo Chick Sex Differentiation", $650,000 ($101,360 to Choi), Oct 2026–Sep 2030.

Georgia Peanut Commission

- PI, "Night Owl: A Low-Cost Smart Drone System for Defending Peanut Farms from Nighttime Wildlife Intrusion", $25,000, Jul 2025–Jun 2026.

USDA (through SARE)
- PI, "MoCoBot: Developing a Low-Cost Night-time Mollusk Control Robot for Strawberry Growers", Graduate Student Grants, $21,964, Sep 2024–Aug 2026.

Assurant
- PI, "Help Desk Call Recording Analysis", $35,000, Jan 2026–Jun 2026.
- PI, "Integrated Help Desk Recording-Ticket Analysis and Deployment Readiness", $40,000, Sep 2026–Feb 2027

TEACHING EXPERIENCE

KSU, Marietta, GA, USA
Instructor
- IT 7143: Cloud Analytics **Sp 2024, Sp 2025, Sp 2026**
- IT 7133: Enterprise AI Applications **Su 2025**
- IT 7103: Practical Data Analytics **Fa 2023, Fa 2024, Fa 2025, Su 2026**
- IT 4773: Machine Learning for Enterprise **Fa 2023, Fa 2024**

ASU, Tempe, AZ, USA
Teaching Assistant
- CSE 450/551: Design Analysis of Algorithms/Foundations of Algorithms: **Jan 2018 – May 2018**
  - Instructor: Dr. Andréa Richa
  - Graded exams, and held office hours (2 hours/week) for tutoring students.
- CSE 310: Data Structures and Algorithms: **Aug 2017 – Dec 2017**
  - Instructor: Dr. Andréa Richa
  - Taught recitation sessions (4 hours/week), graded exams, and provided instructions for C++ programming assignments.
- CSE 205: Object-Oriented Programming & Data Structures **Jan 2016 – May 2016**
  - Instructor: Dr. Xuerong Feng
  - Graded exams, and tutored students for Java programming (4 hours/week).
- CSE 100: Prin. of Programming with C ++ **Jan 2016 – May 2016**
  - Instructor: Dr. Phillip Miller
  - Supervised C++ programming laboratories (5 hours/week), and held tutoring hours (4 hours/week).
- CSE 424: Capstone Project II **Aug 2015 – Dec 2015**
  - Instructor: Dr. Debra Calliss
  - Supervised each project group with their short-term and long-term goals, and graded IT ethics essays.

MENTORING

KSU, Marietta, GA, USA
- Cameron Redovian (Ph.D. in Computer Science) **Aug 2024 – Present**
- Zularbine Kamal (Ph.D. in Computer Science) **Aug 2026 – Present**
- Carter Corbin (M.S. in Intelligent Robotic Systems) **Jun 2026 – Present**
- Shema Nkindi Giscard (M.S. in Computer Science) **Aug 2026 – Present**
- Subeen Choi (B.S. in Industrial Systems Engineering) **Jan 2026 – Present**
- Robel Mamo (M.S. in Computer Science) **Aug 2024 – May 2026**

UCD, Davis, CA, USA
- (Team 1) Satyam Goyal, Rohan Sheth, Akash Seelam, and Kavya Sasikumar
  (Team 2) Grisha Bandodkar, Athul Krishna Sughosh, Sahil Singh, and Shyam Agarwal (B.S. in Computer Science) **Jun 2023 – Feb 2024**
  - Serving as a mentor for the EAAI-24 Mentored Undergraduate Research Challenge

  - Advising in designing a problem, producing a solution and evaluation results, and writing a paper to submit.
- Grisha Bandodkar, Zifei Cheng, Chonghan Wang, and Satyam Goyal (B.S. in Computer Science) **Jan 2023 – Aug 2022**
  - Directing to generate and extend DAVIS-Ag—a public image dataset of simulated plants—with labels, such as instance segmentation, bounding box, camera pose, etc., in a desirable format.
  - Advising in designing reasonable heuristic methods for active vision in agricultural environments and in training a fruit detector for validation of the DAVIS-Ag dataset.
  - Guiding while exploring embedded AI platforms like AI Habitat and AI2-THOR
- Avishai Halev (Ph.D. in Applied Mathematics) **Sep 2022 – Dec 2022**
  - Working with food scientists on efficient parameter search to build a realistic oxidation model fitting empirical datasets.

UoL, Lincoln, UK
- Owen Would (M.Sc. in Robotics & Autonomous Systems) **Mar 2021 – Sep 2021**
  - Advised in studying on deep neural network-based visual anomaly detection of strawberry images particularly while designing a GAN-based method for occluding environments.

ASU, Tempe, AZ, USA
- Sehyeok Kang (M.S. in Computer Engineering) **Mar 2019 – May 2020**
  - Mentored to implement Remote Teammate Localization on a physical robot platform, *Thymio*, while he was working on his thesis on correlation between prediction accuracy and observational information.
- Ricardo Weir (B.S. in Computer Science) **Mar 2018 – Dec 2018**
  - Guided to develop a YOLO-based object-detection pipeline—from data annotation to model validation—to perform automated tracking of individual *Harpegnathos* ants from high-quality video recordings.

MEDIA COVERAGE

- Semafor (Sep 2025) – “It’s common scents”
- WSB Radio (Sep 2025) – “Kennesaw State researchers develop ‘Electronic Nose’ to detect food-borne illness”
- Fox 5 Atlanta (Sep 2025) – “KSU researcher developing AI-powered ‘electronic nose’”
- The Georgia Sun (Sep 2025) – “A KSU Researcher is Creating an Electronic Nose to Sniff Out Foodborne Illness”
- KSU News (Sep 2025) – “KSU researcher developing electronic nose to sniff out foodborne illness”
- KSU News (Jun 2025) – “Kennesaw State researchers developing AI-powered drones to protect peanut farms from wildlife”
- Wisconsin Ag Connection (May 2025) – “Night Vision Robot Protects Strawberry Crops”
- WSB Radio (May 2025) – ‘Kennesaw State University researchers develop AI robot for pest control”
- KSU News (May 2025) – “Kennesaw State researcher develops AI robot to aid farmers in fight against pests”

PROFESSIONAL SERVICE

KSU Research Champion **Aug 2025 – Present**
- Co-leading the Applied Technologies Research Community, promoting collaborative research across the campus.

Workshop Organizer
- ICRA2023: “TIG-IV: Agri-Food Robotics From Farm To Fork” **Jun 2023**

– Full-day workshop on robotic innovations for agri-food systems—from farming to post-harvest processing, cooking, delivery, serving, and legislation

Conference/Journal Reviewer
- IJRR, RAM, RA-L, Biosystm. Eng., Comput. Electron Agric., IROS 2026 (Associate Editor), IROS 2025 (Associate Editor), IROS 2023, ICRA 2023, ICRA 2022, IROS 2022, ICRA 2020

Grant Reviewer
- Research Grant at ASU GPSA **Aug 2017 – May 2018**
- Travel Grant at ASU GPSA **Aug 2016 – Jul 2017**

Conference Session Chair
- TAROS 2021 **Sep 2021**
- CASE 2017 **Aug 2017**

AWARDS

KSU
- Apex Award for research and scholarly excellence **Dec 2025**

KSU PechaKucha Night
- Best Demonstration Award **Nov 2024**

IROS2023 Workshop on Agricultural Robotics for a Sustainable Future
- 2nd Prize for Innovation in Precision Agriculture **Oct 2023**

SWARM 2021
- Best Paper Award **Jun 2021**

ASU Graduate College
- Completion Fellowship ($8,550 plus tuition for 1 credit hour) **Aug 2020**

ASU Ira A. Fulton Schools of Engineering
- Engineering Graduate Fellowship ($700) **May 2020**

ASU School of Computing, Informatics, and Decision Systems Engineering
- Doctoral Fellowship ($4,000) **Mar 2020**

ASU Social Insect Research Group
- Student Research Grants ($1,550) **Nov 2018**
  - Project: Deep Features for Generalizable Insect-behavior Learning.

SSU College of Information Technology
- Bronze Award at Software Development Competition **Oct 2012**
  - Social Alarm: Smart Android Alarm Application (Photos & Demo)

WORK EXPERIENCE

Atlassian, Mountain View, CA
Data Scientist Intern **May 2018 – Aug 2018**
- Jira Duplicate Ticket Detection
  - Designed a deep learning pipeline for human natural language to classify semantically similar tickets from customers.
  - Gathered >124K examples to implement, train, fine-tune, and validate specialized LSTM models.

- Demonstrated 1) significantly higher accuracy than traditional machine learning models, 2) generalizability to the data from different sources of ticket, and 3) feasibility of similarity-based ranking scenarios.

HARDWARE AND SOFTWARE SKILLS

Data Science & Machine Learning:
- PyTorch, TensorFlow, TensorBoard, OpenAI Gym, OpenCV, Scikit-learn, SciPy, NumPy, Pandas, Matplotlib, etc.

Programming Languages:
- Python, Java, C, C++, UNIX shell scripting, GNU make, MySQL, etc.

Operating Systems:
- Microsoft Windows family, Apple OS X, Linux, and other UNIX variants

Others:
- Unity 3D, MATLAB, LaTeX, Git, Android, and TCP/IP programming